CAR SALES IN USA 2014-2015-DATA ANALYTICS PROJECT

CAR SALES IN USA 2014-2015 QUESTIONS

- Which top 5 brands/states sold most cars from 2014-2015 in USA and generated more revenue?
- Which bottom 5 brands/states sold the least cars from 2014-2015 in USA and generated less revenue?
- Which seller or dealership sold most cars.
- Which dealership made huge revenue in sales of cars.
- Which dealership made less money in revenue in sales of cars

DATA SOURCE

- **Is the data valid, credible, accurate and is it from a trusted data source?**
  The data is from Kaggle and it is not biased in any way and it has been cleaned and verified to avoid/prevent any errors during the analysis process and also so that the insights are accurate and correct.

DATA CLEANING

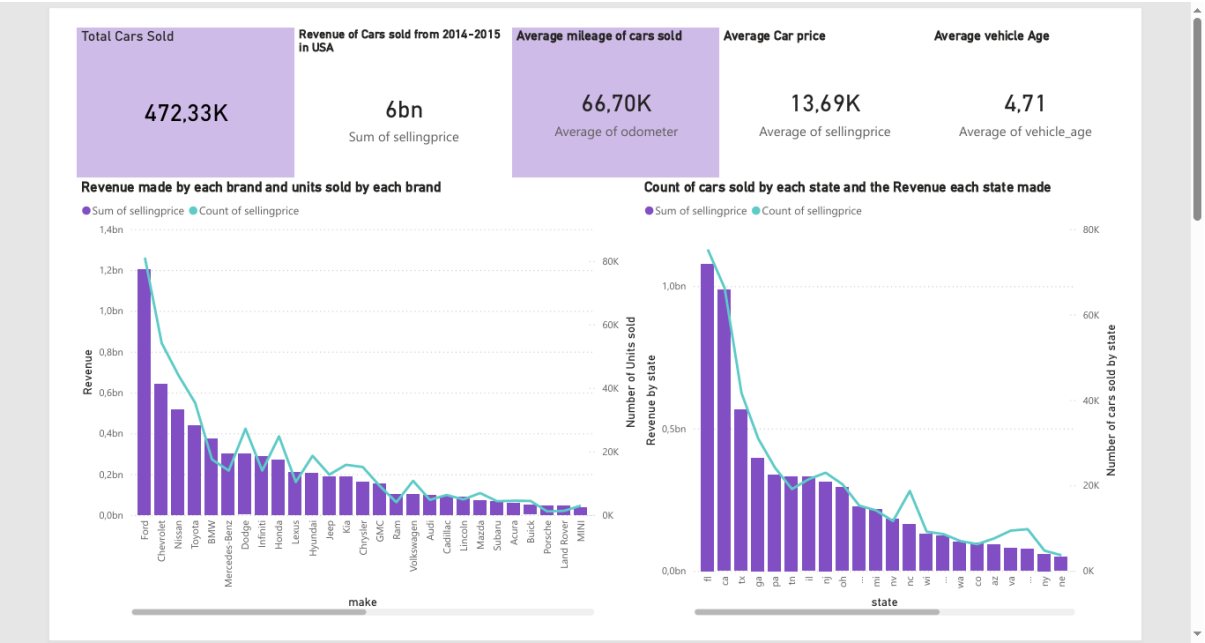
- Data was Cleaned in both Microsoft excel and Python by checking if there are duplicates in the dataset, removing null records, fixing typos and also checking the validity of data in the dataset and removing data that is not valid.

LIMITATIONS AND PROBLEMS

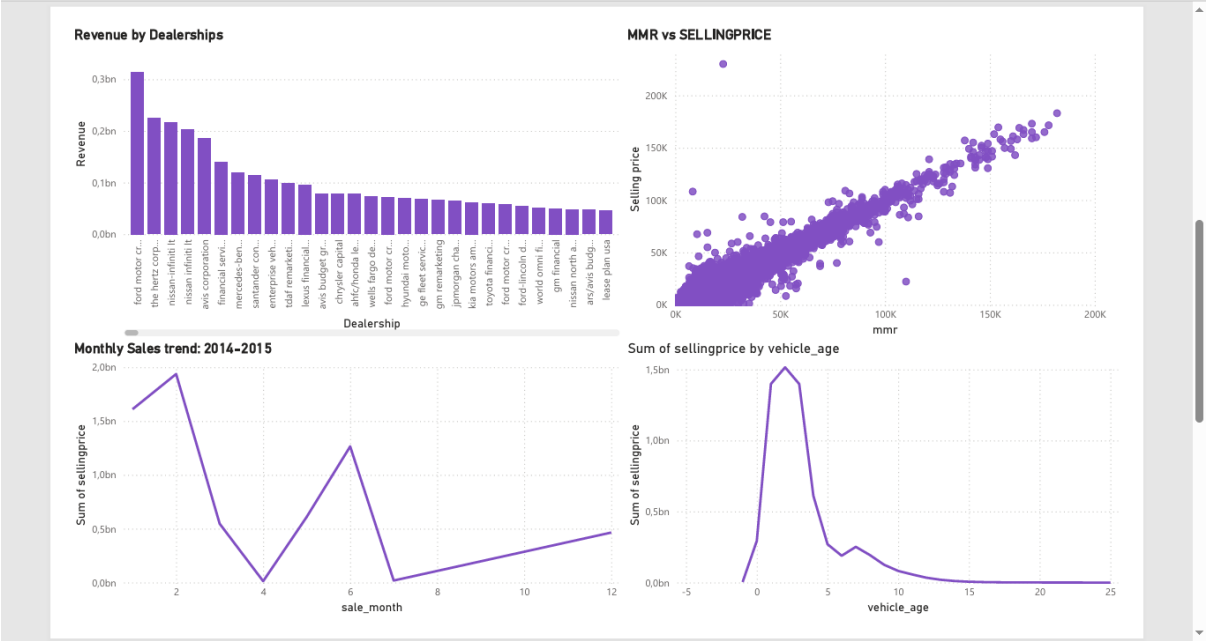
- Due to the specs of my PC, I had a problem in cleaning the dataset on MySQL as the data file was too large and heavy for my PC and it took time for the data to be imported on MySQL. To solve this issue, I utilized Python to clean my data as it is faster and more efficient in importing dataset.

FINDINGS AND INSIGHTS

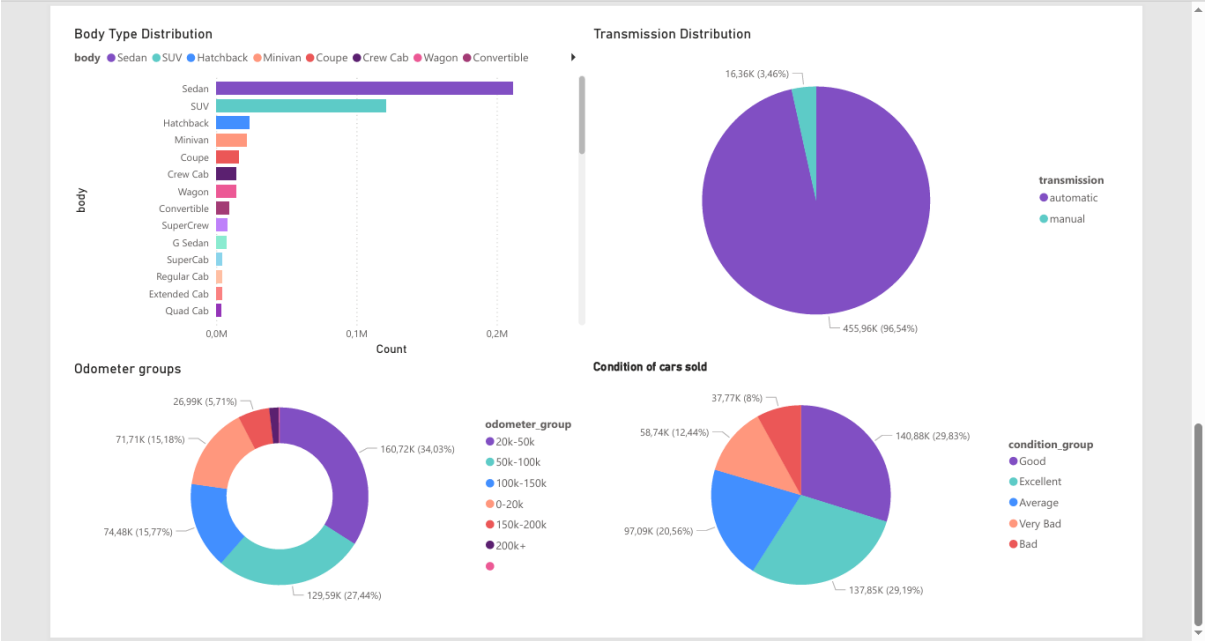
Upon processing and analysing the data about sales of cars in USA from 2014 till 2015, I discovered the following:



- 472,33k cars were sold from 2014 till 2015 in USA
- The sales of cars amounted to a revenue of $6 billion
- The average price of a car was estimated to $13,69k
- Ford, Chevrolet, Nissan, Toyota, and BMW were the top sold cars and they generated more revenue.
    - My assumption for the first two leading brands which are Ford and Chevrolet is that these brands were leading mainly because they are locally produced and they originated from the USA which make these two leading brands to be highly purchased by citizens, private sectors and government entities.
- Luxurious cars, to mention few, Lamborghini and Ferrari didn't generate much revenue and sold less units.
    - My assumption is that these luxurious cars are expensive and are not recommended for day for day use because they are expensive to maintain, that's why they are in the top ten of least bought cars and thereby didn't generate that much revenue.



- Florida, California, Texas, Georgia and Pennsylvania are the top 5 states which generated more revenue in sales of cars and thereby selling large amounts of cars.
- Ford Motor credit company LLC, the hertz corporation, Nissan Infiniti LT were the top 3 car selling dealerships and this is justified by the top 5 selling brands which are Ford, Chevrolet, Nissan, Toyota and BMW
- Manheim Market Report (MMR) which is the industry market price for the vehicles correlated with the selling price of vehicles as seen on the scatterplot. This suggest that the was a positive correlation as most cars sold, their MMR value was less than the selling price which means that there was profit in the sales. Even though there was a huge positive correlation in MMR and selling price, there were some outliers in the scatterplot (e.g. MMR = $22800 & selling price = $230000) which means that some sales were unusual whereby the MMR value was way less than the selling price indicating huge profits and in some case the selling price was way less than the MMR value which indicated loss in revenue.
- January and February were the months were the sales of cars were at the peak level. The sales started to drop from march and they were the lowest in April and July. From August till December the sales exponentially increased.
- Cars with age more than 5 years generated less revenue as seen on the line chart. This indicates that individuals and companies spent more money on new/recently released vehicles.



- Sedans were leading the charts with 200k+ sedans sold during the 2014-2015 period followed by SUV's with 100k+
- 96,54% of cars sold consist of automatic transmission while the remaining 3,46% consist of manual transmission.
    - My assumption is that most people preferred the automatic transmission as opposed to manual transmission because automatic transmission is more efficient and easier than manual transmission and mostly people preferer hassle free options.
- Most cars sold had a less mileage in the category of 20k-50k. This most of the times goes hand in hand with the age of the vehicle. The age of the vehicle and the mileage data correlates because cars with less than 5 years of age generally have less kilos/miles in mileage while cars with age more than 5 years have high mileage and generally people prefer newer cars with less mileage as seen on the charts that newer cars with less mileage generated more revenue as opposed to cars with high age and high mileage.
- In the category of the condition of the cars sold. The "Good" and "Excellent" categories lead the charts and this category also correlates with the mileage and age of the vehicle. This is justified because mostly newer/recently released cars tends to have less mileage and the conditions of these cars tends to be good or excellent.